

C3

The frontier, the case, and the whole course

~55 min · what is moving · a board that could not act · six weeks in one sentence

BY THE END OF THIS CAPSULE YOU WILL BE ABLE TO

Argue what a governance structure must actually do to stop something, and state in one sentence what these six weeks were about.

What is moving — and who will settle it

Pre-read 5 leaves it unresolved. No expert settles this — only whoever holds the controls.

REASONING MODELS

Compute moved from training into inference — the model thinks for longer at answer time. Your cost model from Week 5 changes shape: you are now buying thinking, per request.

What to watch: cost per **SUCCESSFUL** task, not per token.

MULTIMODAL AND AGENTIC

One model over text, images, audio and screens, wired to tools. Week 4's agent loop is the primitive; the reliability arithmetic does not improve because the model got better at seeing.

What to watch: the number of steps, not the quality of one.

PHYSICAL AI

The C1 fourth column, arriving. No review step, no undo, a safety case owed BEFORE deployment rather than a monitor after it.

What to watch: who signs the safety case.

HOW TO KEEP LEARNING AFTER THIS

Model cards and system cards over launch posts • the benchmark's own paper over the bar chart • a regulator's text over a summary of it • and build the small thing yourself.

The habit: read what was measured.

Takeaway The facts expire; the questions do not — and the argument is settled by whoever holds the controls.

CASE • GOVERNING OPENAI (A)

Tuesday, 21 November 2023.

Four days ago the board of the most valuable AI company in the world removed its chief executive. Tonight almost every employee has threatened to resign. You have read the brief. Do not expect me to re-narrate it.

Who controls this company?

The control lines are real. So is everything standing outside them.



Takeaway Control is the ability to withhold something that cannot be replaced quickly.

Five days — the dates, and the four numbers

WORKED

The compression is a fact about governance, not a detail of the story.

When	What happened	The number
Fri 17 Nov 2023	Board removes Altman by video call. Brockman removed as chair, resigns hours later.	6 directors on the board — 4 of them decide
Sun 19 Nov 2023	Talks to reinstate him fail. Microsoft announces it will hire Altman and Brockman.	US\$13bn committed by one shareholder with no board seat
Mon 20 Nov 2023	Employee letter threatening to resign, signed in waves through the day.	more than 700 signatures of about 770 staff
Tue 21 Nov 2023	Altman returns. The board that removed him is replaced.	US\$86bn — the tender-offer valuation then in flight
Elapsed	From the removal call to the reinstatement announcement.	about 106 hours

Takeaway Every formal power sat with the six. Every practical power sat outside them.

Formal rights and actual power

The left column is in the documents. We build the right one together, out loud.

FORMAL RIGHTS — what the documents grant	ACTUAL POWER — what could be withheld
The board may remove the CEO, without cause, by majority	Employees can leave — and take the only asset with them
The board owes its duty to the mission, not to shareholders	Microsoft can supply, or not supply, the compute
The non-profit controls the for-profit through the GP	A shareholder can offer 770 people a landing place in 48 hours
The board may unilaterally declare AGI and end Microsoft's licence	Investors can withdraw a tender offer that is days from closing
Four directors were legally sufficient on 17 November	None of the right-hand column appears in any governance document

WilmerHale later found the prior board “acted within its broad discretion” — so the left-hand column was never in doubt.

Takeaway The two columns do not overlap. That is the case, and it is not unique to OpenAI.

(B) — what the record says, and what it does not

The independent review of 8 March 2024, and the account published eleven weeks after it.

WilmerHale, after dozens of interviews and 30,000 documents: the removal arose from “a breakdown in trust”, and “did not arise out of concerns regarding product safety”.

That says safety was not the REASON. It does not say the concerns were untrue: a breakdown in trust is entirely consistent with them being well founded.

Then Helen Toner, one of the removing directors, 28 May 2024: the board learned of ChatGPT’s launch from Twitter, and Altman owned the Startup Fund while presenting as a director with no financial interest. Taylor and Summers rejected her account, resting on the WilmerHale finding.

Both can hold: their rebuttal is about the NEW board, her allegations concern the PRIOR one — and nothing here actually contradicts.

Takeaway “Not the reason” and “not true” are different claims. Most disputes you will arbitrate look like this.

What the structure became, 2024–2026

The board lost. Then, over two years, most of what it wanted was built — and some of it was removed.

	What changed	
THE GATE ARRIVED	Oct 2025 — the recapitalisation and the Attorneys-General MOU. The OpenAI Foundation appoints and can replace every director; a Safety and Security Committee may halt a model release; no expiry.	FOR: the board of 2023 had no such lever.
THE MISSION THINNED	The word “safely” disappears from the stated mission, and on 27 Apr 2026 the Charter is replaced by “Our Principles” — merge-and-assist and “primary fiduciary duty is to humanity” both gone.	AGAINST: the thing being protected got smaller.

Also in the two years between: the profit cap abolished, Microsoft holding ~27%, and the AGI carve-out now verified by an independent expert panel.

Takeaway A stronger institution, or a better-documented retreat. Not settled — and you will find out.

THE CASE • ONE LINE

The binding constraint was governance, not technology.

Nothing in this case turned on a model, a benchmark or an engineering decision. It turned on who could withhold what, and on whether anyone had written that down. Hold that sentence — and let us look at all six weeks.

What six weeks actually was

COUNTED

Counted from the course folder, not estimated. Every number below is something you did.

	How much	
Sessions • contact hours	6 • 18 hrs	plus 68.5 total planned learning hours
Pre-read briefs • pages	37 • 191 pages	16 pages in Week 1 • 51 in Week 6
Case studies	7	Moderna • DBS • Med-PaLM • AI Wars • Anthropic • Ant • OpenAI
Hands-on builds • notebooks	7 • 9	classifier • tokeniser • RAG • evals • agent • cost • red team
Polls, quizzes and exercises	about 80	you answered before you were told, every single week
Regimes • assessments • slides	5 • 3 • 508	three graded assessments • 508 slides across the six sessions
You	28 teams • 172	from every function, which is the point

Takeaway You did not take a survey course. You built seven things and read a hundred and ninety pages.

Six weeks, one argument

Read the right-hand column down. That is the course.

Week	What you built	What it was actually teaching
1 • The jagged frontier	A discriminative classifier against a general-purpose API	That “AI” names four different contracts, and the cheap one is often correct
2 • Data and the machine	A tokeniser, and a RAG pipeline over your own documents	That the model is rentable and your data is not
3 • Making it behave	Structured output, plus a ten-case evaluation harness	That you cannot manage what you have not defined a measurement for
4 • Agents	An agent with five tools, and its failure trace	That reliability compounds downward, and an early error becomes a premise
5 • The economics	A cost-to-serve calculator for your own use case	That inference is a per-request cost line, and training was capital
6 • Responsibility	A working attack against your own build, and the guardrail that closed it	That the constraint is never the algorithm

Takeaway One sentence: decide what kind of problem you have, then price it, then govern it.

Five threads, running through all six weeks

Every one of them started in Week 1 and landed today.

Thread	Weeks	What it means when you are the one deciding
① Paid once vs paid forever	W1 · W5	Training is capital. Inference is a cost line that scales with success — the more it works, the more it costs.
② The model is rentable. Your data and your governance are not.	W2 · W6	Anyone can call the same API. Nobody else has your corpus, your consent basis or your approval chain.
③ A gate someone can actually operate	W3 · W4 · W6	Human-in-the-loop is a specification — window, evidence, authority — or it is a word in a slide deck.
④ You cannot test what you did not imagine	W3 · W6	Ten cases you wrote beat a benchmark you did not. And the attack you ran today was the case you had not written.
⑤ The constraint is never the algorithm	W1 → W6	Moderna, DBS, Med-PaLM, Ant Group, OpenAI. Five cases, five bottlenecks, not one of them a model.

Takeaway If you keep five things from this course, keep these. They outlive every model version.

A2 — every deliverable, and the week it came from

Nothing in A2 is new work. It is four things you have already built, written up.

A2 deliverable	Where it came from	What to reopen tonight
Tool-use design and the agent loop	Week 4	Your agent notebook and its failure trace
Cost-to-serve estimate	Week 5	Your cost calculator — change the volume, not the method
Guardrail checklist	TODAY	Pre-read 1's failure grid + the six-category draw from C2·B
The trade-off you made, argued	Rubric 1, criterion 3	C1's two-question diagnostic: visible? undoable?

The strongest A2s name a trade-off and defend it. The weakest list every control and commit to none.

Takeaway If you attended today, the guardrail section is already written — you wrote it in C2·B this evening.

The dates, unambiguously

This is the last physical class. Week 7 is a submission window with no teaching.

What	When	Note
A2 — GROUP report	Sunday 20 September 2026, 23:59 SGT	20% • ONE submission per team, and the whole team shares the mark
Peer rating	Thursday 24 September 2026, 23:59 SGT	A participation requirement, not optional — a signal, never a multiplier
End-of-course project — INDIVIDUAL	Sunday 4 October 2026, 23:59 SGT	44% • one submission each • repo + recorded demo + trade-off report
A2 feedback	Returned online	⚠ THERE IS NO WEEK-7 CLASS. Feedback is offline; questions by email or forum

Takeaway Three dates, all 23:59 Singapore time. Put them in your phone before you leave this room.

Where your grade stands, right now

WORKED

Two of the four are behind you. The largest one is not.

Component	Weight	Status
A1 — Foundations Build • individual	16%	DONE
Class participation	20%	DONE today • one action left, the peer rating on 24 Sep
A2 — Applied AI System • GROUP	20%	LIVE • due Sun 20 Sep
End-of-course project • INDIVIDUAL	44%	LIVE • due Sun 4 Oct

36% is settled. The remaining 64% splits 20 / 44 — which is why the project, not A2, is where the marginal hour goes.

Takeaway 36% is done. 64% is still ahead — and 44 of it is one deliverable, due on 4 October.

CAPSULE 3 • DONE

That is the course.

Six weeks, seven builds, seven cases, five threads. Three dates. One last thing to say, and it is not about the assessment.